

14. Pure bending of beam will have
 A) Tensile and shear stresses
 B) Compressive and shear stresses
 C) Tensile and compressive stresses
 D) Torsional and shear stresses
15. One of the assumptions made in the theory of pure bending is the radius of curvature is _____ the dimensions of the cross-section.
 A) larger than
 B) smaller than
 C) equal to
 D) not comparable to
16. In I-Section of a beam subjected to transverse shear force, the maximum shear stress is developed
 A) At the centre of the web
 B) At the top edge of the top flange
 C) At the bottom edge of the top flange
 D) At the junction of flange and web
17. Which of the following is a differential equation for deflection?
 A) $dy/dx = (M/EI)$
 B) $dy/dx = (MI/E)$
 C) $d^2y/dx^2 = (M/EI)$
 D) $d^2y/dx^2 = (ME/I)$
18. The vertical distance between the axis of the beam before and after loading at a point is called as ____
 A) deformation
 B) deflection
 C) slope
 D) bending
19. _____ is a measure of the strength of shaft in rotation
 A) Torsional modulus
 B) Section modulus
 C) Polar modulus
 D) Torsional rigidity
20. Calculate the torque, if the diameter of the shaft is 50 mm and revolutions are at 130 rpm. The maximum shear stress is 62.5 N/mm^2 .
 A) 1764 Nm
 B) 1378 Nm
 C) 1534 Nm
 D) 1294 Nm

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) A tapering bar has depth ' b_1 ' at one end and it tapers uniformly to a depth ' b_2 ' at the other end with uniform thickness ' t ' for a length of ' L '. If modulus of elasticity of the material is ' E ', Determine its change in length when subjected to an axial force ' P '.
 b) A 500 mm long bar has rectangular cross-section $20\text{mm} \times 40\text{mm}$. This bar is subjected to:
 (i) 40 kN tensile force on $20 \text{ mm} \times 40 \text{ mm}$ faces,
 (ii) 200 kN compressive force on $20 \text{ mm} \times 500 \text{ mm}$ faces, and
 (iii) 300 kN tensile force on $40 \text{ mm} \times 500 \text{ mm}$ faces.
 Find the change in volume if $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.
2. a) A stepped bar with three different portions has a fixed support at one of its ends. The stepped bar is subjected to forces as shown in Fig.2(a) Determine the stresses and deformations induced in each portions. Also find the net deformation induced in the stepped bar. Take $E=200\text{Gpa}$.

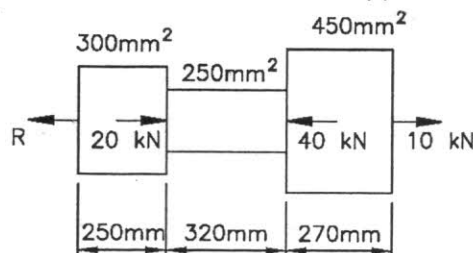


Fig. 2 (a)

- b) Derive the expression for volumetric strain of a thin cylindrical shell subjected to pressure P in terms of Young's Modulus and Poisson's Ratio.
3. a) Draw the stress strain curve for mild steel indicating all salient points on it.
 b) Distinguish between:
 i) Elasticity and Plasticity
 ii) Toughness and resilience.

- c) A cylindrical pressure vessel has inner and outer radii of 200mm and 250mm respectively. The material of the cylinder has an allowable normal stress of 75MN/m^2 . Determine the maximum internal pressure that can be applied and also draw the sketch of radial pressure and circumferential stress distribution.

8 L3 2 2

Unit – II

4. A simply supported beam AB of 6m span is loaded as shown in Fig. 4(a). Draw the shear force and bending moment diagram. Also indicate the point of contra flexure if any.

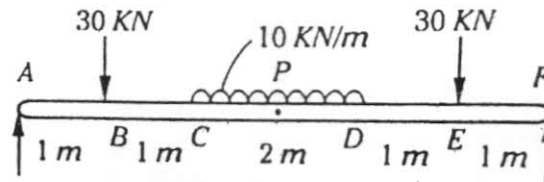


Fig.4(a)

16 L3 3 2

5. a) Define Bending moment and Shear force diagrams. Discuss the sign conventions used for shear force and bending moments.
 b) Explain types of supports used in beams.
 c) Show that for a rectangular cross section, shear stress distribution varies parabolically across depth. Further show that maximum shear stress is 1.5 times average stress.
6. a) A beam of T section having flange dimensions $100\text{mm} \times 20\text{mm}$ and web dimensions $80\text{mm} \times 20\text{mm}$ is used as a simply supported beam over a span of 8 meters. It carries a uniformly distributed load of 1.5 kN/m over its entire span. Calculate the maximum tensile and compressive stresses occurring in the section.
 b) A beam of an "I" section $200\text{mm} \times 300\text{mm}$ has web thickness 10mm and flange thickness 10mm as shown in Fig. 6(b). It carries a shearing force of 10kN at a section. Sketch the shear stress distribution across the section.

6 L1 3 1

2 L1 3 1

8 L2 4 1

8 L3 4 2

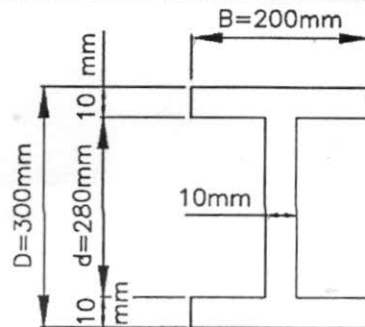


Fig. 6(b)

8 L3 4 2

Unit – III

7. a) Derive an expression for deflection for a simply supported beam subjected to UDL.
 b) Find the deflection at the free end of the cantilever shown in Fig. 7(b), The flexural rigidity $EI = 80 \times 10^4\text{ kN-m}^2$.

8 L2 5 1

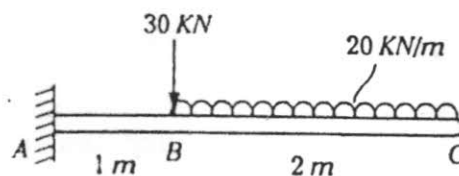


Fig. 7(b)

8 L3 5 2

8. a) A solid shaft rotating at 500 rpm transmits 30 kW. Maximum torque is 20% more than the mean torque. Material of the shaft has the allowable shear stress of 65 MPa and Modulus of rigidity 81 GPa. Angle of the twist in the shaft should not exceed 1° in one-meter length. Determine the diameter of the shaft.
- b) Derive the relationship between torque and shear stress in a solid circular shaft. Also mention assumptions made.

8	L3	5	2
8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
